



SHREE SWAMINARYAN INSTITUTE OF TECHNOLOGY

Near Agora Mall & Indira Bridge, Bhat Circle, Ahmedabad
Airport-Gandhinagar Highway, Bhat, Gandhinagar-382428

CE/IT Department

Subject: DBMS (3130703)

Sem. 3th

Questions Bank

Q.1	Explain detailed 3-level architecture of database system with diagram. Or Explain three levels of data abstraction. Or Define following terms. 1) Schema 2) Database Management System 3) Physical Data Independence
Q.2	What is DBA? What are the main duties of DBA? Or What is the main role of DBA? Or Describe tasks performed by the Database Administrator
Q.3	Explain various database users and administrator. Or What is the difference between Database user and Database administrator?
Q.4	What are the advantages and disadvantage of database over file system?
Q.5	Draw E-R Diagram of following scenario. Make necessary assumptions and clearly note down same. -- College manually operated library to fully computerized. OR Construct E-R diagram for a hospital with a set of patients and medical doctors. Associate with each patient a log of various tests and examinations conducted. OR Draw E-R diagram for student management system with the necessary assumption. OR Draw an E-R diagram of following scenario. Make necessary assumptions and clearly note down the same. We would like to make our College's manually operated Library to fully computerized .
Q.6	What is constraint? What are its types? Explain.
Q.7	Explain joins with suitable example.
Q.8	What is relational algebra? Explain select and project operations with example.
Q.9	What is Functional Dependency? Explain Types of FD.or What is functional dependency? Explain its usage in database design.
Q.10	Explain Armstrong's Axioms for FD.
Q.11	Compute the closure of the following set F of functional dependencies for relational schema $R = (A, B, C, D, E, F)$: $F = (A \rightarrow B, A \rightarrow C, CD \rightarrow E, CD \rightarrow F, B \rightarrow E)$.
Q.12	We have following relations: Supplier(S#,sname,status,city)



SHREE SWAMINARYAN INSTITUTE OF TECHNOLOGY

Near Agora Mall & Indira Bridge, Bhat Circle, Ahmedabad
Airport-Gandhinagar Highway, Bhat, Gandhinagar-382428

CE/IT Department

Subject: DBMS (3130703)

Sem. 3th

	Parts(P#,pname,color,weight,city) SP(S#,P#,quantity) Answer the following queries in SQL. 1. Find total number of city of all suppliers. 2. Find name of supplier for city = 'Delhi'. 3. Find s# of supplier who supplies 'red' part.
Q.13	Explain the purpose and application of DBMS .
Q.14	What do you mean by integrity constraints? Discuss various integrity constraints. Or What is integrity constraint? Explain primary key, reference key and check constraint with SQL syntax.
Q.15	Define: (1) Primary key (2) Foreign key (3) Unique Key (4) Not null (5) Candidate key (6) Super Key or Define the terms: a) Primary Key b) Super Key or Define the terms : a) foreign key b) candidate key
Q.16	List and explain various data models used for database design.
Q.17	Explain following relational algebra operation (i) Natural join operation (ii) Selection and projection operation or What is Relational Algebra? Define Relational Algebra Operation cross product with example or List the type of joins in relational algebra. Explain with example
Q.18	Consider following schema and represent given statements in relation algebra form. • Branch(branch_name,branch_city) , • Account(branch_name, acc_no,balance) • Depositor(Customer_name, acc_no) 1. Find out list of customer who have account at 'Baroda HO' branch. 2. Find out all customers who have account in 'Ahmedabad' city and balance is greater than 10,000. 3. Find out list of all branch names with their maximum balance. OR Write Relational Algebra syntax for the following queries. Employee(eno,ename,salary,designation) Customer(cno,cname,address,city) 1) Find out name of employees who are 'Manager'. 2) Display name of customers. 3) Retrieve Employee records whose salary is less than 20,000 OR Write Relational Algebra syntax for the given queries using the following database. Employee(eno,ename,salary,designation) Customer(cno,cname,address,city) 1) Find out name of employees who are also customers. 2) Find out name of person who are employees but not



SHREE SWAMINARYAN INSTITUTE OF TECHNOLOGY

Near Agora Mall & Indira Bridge, Bhat Circle, Ahmedabad
Airport-Gandhinagar Highway, Bhat, Gandhinagar-382428

CE/IT Department

Subject: DBMS (3130703)

Sem. 3th

customers. 3) Display all names who are either employees or customers.

OR

Consider the relational database given below. Give an expression in the relational algebra to express each of the following queries: Employee (person-name, street, city), Works (person-name, company-name, salary), Company (company-name, city), Manages (person-name, manager-name)
(1) Find name of all employees. (2) Find city of employee whose name is 'jashu'. (3) Find name and city of all employees who are having salary > 50000. (4) Find total salary of all employees who are working for company 'HCL'.

OR

Consider the following relational database schema consisting of the four relation schemas:
passenger (pid, pname, pgender, pcity) agency (aid, aname, acity) flight (fid, fdate, time, src, dest) booking (pid, aid, fid, fdate) Answer the following questions using relational algebra queries.
a. Get the details about all flights from Chennai to New Delhi. b. Get the complete details of all flights to New Delhi. c. Find the passenger names for passengers who have bookings on at least one flight

OR

TABLE Worker(WORKER_ID INT NOT NULL PRIMARY KEY, FIRST_NAME CHAR(25), LAST_NAME CHAR(25), SALARY INT(15), JOINING_DATE DATETIME, DEPARTMENT CHAR(25));

TABLE Bonus(WORKER_REF_ID INT, BONUS_AMOUNT INT(10), BONUS_DATE DATETIME, FOREIGN KEY (WORKER_REF_ID), REFERENCES Worker(WORKER_ID));

TABLE Title(WORKER_REF_ID INT, WORKER_TITLE CHAR(25), AFFECTED_FROM DATETIME, FOREIGN KEY (WORKER_REF_ID) REFERENCES Worker(WORKER_ID));

Consider above 3 tables, assume appropriate data and solve following SQL queries

1. Find out unique values of DEPARTMENT from Worker table
2. Print details of the Workers whose SALARY lies between 100000 and 500000.
3. Print details of the Workers who have joined in Feb'2014.
4. Fetch worker names with salaries >= 50000 and <= 100000.

OR

TABLE Worker(WORKER_ID INT NOT NULL PRIMARY KEY, FIRST_NAME CHAR(25), LAST_NAME CHAR(25), SALARY INT(15), JOINING_DATE DATETIME, DEPARTMENT



SHREE SWAMINARYAN INSTITUTE OF TECHNOLOGY

Near Agora Mall & Indira Bridge, Bhat Circle, Ahmedabad
Airport-Gandhinagar Highway, Bhat, Gandhinagar-382428

CE/IT Department

Subject: DBMS (3130703)

Sem. 3th

	CHAR(25)); TABLE Bonus(WORKER_REF_ID INT,BONUS_AMOUNT INT(10),BONUS_DATE DATETIME,FOREIGN KEY (WORKER_REF_ID),REFERENCES Worker(WORKER_ID)); TABLE Title(WORKER_REF_ID INT,WORKER_TITLE CHAR(25), AFFECTED_FROM DATETIME,FOREIGN KEY (WORKER_REF_ID)REFERENCES Worker(WORKER_ID)); Consider above 3 tables ,assume appropriate data and solve following SQL queries 1. Print details of the Workers who are also Managers. 2. SQL query to clone a new table from another table. 3. Fetch the list of employees with the same salary. 4. Fetch “FIRST_NAME” from Worker table in upper case.
Q.19	Explain following relation algebra operations: (i) full outer join (ii) projection (iii) Cartesian product.
Q.20	(i)What is DBMS? (ii)What is E-R Model?
Q.21	(i)Differentiate between Disjoint and Overlapping Constraints (ii) Explain the difference between a weak and a strong entity set. Or What is weak entity? How the weak entity can be converted to the strong entity? Show the symbol for representing weak entity or Differentiate strong entity set and weak entity set. Demonstrate the concept of both using real-time example using E-R diagram
Q.22	Explain following terms with reference to E-R model. 1. Entity &it's Attributes. 2. Relationship Type (including recursive). 3. Cardinality & Participation. 4. Weak-Entity Type & Relationship Type
Q.23	Explain specialization and generalization of E-R Diagram with example OR Why do we require E-R model? Explain the term ‘Generalization’, ‘Specialization’ and ‘Aggregation’.
Q.24	Differentiate single-valued and multi-valued attributes with example.
Q.25	What is closure of a Set of Functional Dependencies?
Q.26	Describe Data Definition Language and Data Manipulation Language. Or What is Data Definition Language? List DDL statements and explain anyone with an example.
Q.27	What is the similarity between relational model and E-R model? How the entity, attributes, primary key and relationship are shown in the relational model.



SHREE SWAMINARYAN INSTITUTE OF TECHNOLOGY

Near Agora Mall & Indira Bridge, Bhat Circle, Ahmedabad
Airport-Gandhinagar Highway, Bhat, Gandhinagar-382428

CE/IT Department

Subject: DBMS (3130703)

Sem. 3th

Q.28	Differentiate lossy decomposition and lossless decomposition
Q.29	When Join is used in SQL? Explain Left outer, Right outer and Full outer join with SQL syntax
Q.30	List and explain mapping cardinalities of E-R diagram with example.
Q.31	What are the importance of Primary key and Unique key in database? Explain with example
Q.32	Consider the relation scheme $R = \{E, F, G, H, I, J, K, L, M, N\}$ and the set of functional dependencies $\{ \{E, F\} \rightarrow \{G\}, \{F\} \rightarrow \{I, J\}, \{E, H\} \rightarrow \{K, L\}, K \rightarrow \{M\}, L \rightarrow \{N\} \}$ on R. What is the key for R? Or Consider a relation scheme $R = (A, B, C, D, E, H)$ on which the following functional dependencies hold: $\{A \rightarrow B, BC \rightarrow D, E \rightarrow C, D \rightarrow A\}$. What are the candidate keys of R? (Any 1 in case of more than one candidate key)
Q.33	List and explain ACID properties with respect to Database transaction. Or List and describe ACID property of transactions. Or What is the atomicity and consistency property of transaction?
Q.34	Explain the concept of Conflict Serializable with suitable schedules. Or What is the schedule in truncation? How to identify that the given schedule is conflict serializable? Explain with example
Q.35	List and explain types of locks in transactions.
Q.36	Explain following SQL commands with syntax and significance. (i) Grant & Revoke (ii) Commit & Rollback or Describe Grant and Revoke commands with suitable example.
Q.37	What is a deadlock in transaction? How to detect deadlock in system? Explain with example. Or Explain deadlock with suitable scheduling examples.
Q.38	Write short note on query processing. Or Explain various steps involved in query processing with example. Or What is Query processing? Explain why 'Parsing and translation' and 'Optimization' steps are required for query processing.
Q.39	List the techniques to obtain the query cost. Explain any one.
Q.40	Differentiate shared lock and exclusive lock in lock-based protocol.
Q.41	What is dirty write in the transaction? Explain with example.
Q.42	Describe the two-phase locking protocol in detail. Or What is the use of two-phase locking protocol in concurrency control? Or How does two phase locking protocol differ from timestamp based protocol? Explain timestamp-ordering protocol
Q.43	Describe various state of transaction. Or How does 'partial commit' state differ from 'commit' state of the transaction?



SHREE SWAMINARAYAN INSTITUTE OF TECHNOLOGY

Near Agora Mall & Indira Bridge, Bhat Circle, Ahmedabad
Airport-Gandhinagar Highway, Bhat, Gandhinagar-382428

CE/IT Department

Subject: DBMS (3130703)

Sem. 3th

Q.44	Differentiate closed hashing and open hashing in DBMS.
Q.45	What is the role of an index in the database management system? Explain dense index with example. Or What is index in the database? Explain sparse indices and Dense indices with proper example
Q.46	What is log-based recovery? List and explain various fields use in log records for log-based recovery.
Q.47	Discuss view serializability in transactions.
Q.48	What is the use of a cursor in PL/SQL? Explain with example.
Q.49	Describe two rules of mandatory access control.
Q.50	Explain SQL injection in brief.
Q.51	Write a PL/SQL program that fetches records of all students and insert record as students having CPI > 4 in ELIGIBLE table and students having CPI <= 4 in NOT_ELIGIBLE table from student_master table.
Q.52	What is the limitation of index-sequential file? Explain with example how B+ tree overcomes it.
Q.53	What is the difference between Join and Sub query? Explain any two built-in function for following category. 1) Numeric 2) String 3) Date
Q.54	Differentiate dynamic hashing and static hashing.
Q.55	Enlist and explain user authorization to modify the database schema.
Q.56	Why does the trigger require in database? Write SQL syntax for creating database trigger.
Q.57	When do we require to use group by clause? How aggregate functions are used with group by clause?
Q.58	What is redundancy? Explain insert, update and delete anomalies in database with example. Or What are anomalies in database design? How can we solve it?
Q.59	(1)What is Canonical cover ? (2) What is Normalization ? (3) What are the uses of Functional Dependency ? (4) What is the purpose of normalization ? (5) What is the need for normalization?
Q.60	What is closure of a Set of Functional Dependencies ? or Explain how to find closure of a set of attributes?
Q.61	List the desirable properties of Decomposition.
Q.62	Explain 1NF, 2NF, 3NF and BCNF, 4NF & 5NF. Or What is the advantage of using 3NF? Explain with example. Or Explain with suitable example, the process of normalization covering from 1NF to 3NF. Or Why normalization should be performed on a table and what are its benefits. Or Prove the statement “Every relation which is in BCNF is in 3NF but the converse is



SHREE SWAMINARAYAN INSTITUTE OF TECHNOLOGY

Near Agora Mall & Indira Bridge, Bhat Circle, Ahmedabad
Airport-Gandhinagar Highway, Bhat, Gandhinagar-382428

CE/IT Department

Subject: DBMS (3130703)

Sem. 3th

not true” or

Explain 3NF & BCNF. What is the difference between them?

Q.63 What is Normalization? Normalize below given relation up to 3NF

STUDENT:

Stu ID	Stu Name	City	Pin code	Project ID	Project Name	Course	Content
S101	Ajay	Surat	326201	P101	health	programming	C++, Java,C
S102	Vijay	Pune	325456	P102	social	WEB	HTML,PHP,ASP

Or

A college maintains details of its lecturers' subject area skills. These details comprise: Lecturer Number, Lecturer Name, Lecturer Grade, Department Code, Department Name, Subject Code, Subject Name, Subject Level

Assume that each lecturer may teach many subjects but may not belong to more than one department. Subject Code, Subject Name and Subject Level are repeating fields.

Normalise this data to Third Normal Form.

Or

In the BCNF decomposition algorithm, suppose you use a functional dependency $\alpha \twoheadrightarrow \beta$ to decompose a relation schema $r(\alpha, \beta, \gamma)$ into $r_1(\alpha, \beta)$ and $r_2(\alpha, \gamma)$.

1. What primary and foreign-key constraint do you expect to hold on the decomposed relations?
2. Give an example of an inconsistency that can arise due to an erroneous update, if the foreign-key constraint were not enforced on the decomposed relations above.
3. When a relation is decomposed into 3NF, what primary and foreign key dependencies would you expect will hold on the decomposed.